

# DATA ANALYTICS

## Report

PRESENTED TO

**HLL**

PRESENTED BY

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# Building an Analytics Framework for Horizon Learning Labs Global Growth

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## Vision

As Horizon Learning Labs expands its hubs **data will become one of its most valuable assets**. To sustain growth, ensure consistency, and communicate measurable impact, HLL can evolve from tracking data to indepth data understanding and leveraging.

## Current Landscape

HLL's IT systems and dashboards already provide a solid foundation for visibility: learner progress, attendance, and LMS activity. ***This is the descriptive analytics layer***: it answers what is happening. **HLL's LearnTools** platform already empowers coaches and learners with visual dashboards that help individuals track growth, set goals, and own their learning journey.

The next strategic step could be to consolidate these micro-level data informed insights into a global analytics overview that tracks and analyses metrics across hubs and regions.

By **transforming this data into actionable insights**, HLL can communicate key patterns, trends, and findings to stakeholders, providing a bird's-eye view of company-wide performance. This global perspective links learning outcomes to operational and strategic goals, turning existing data into a powerful guide for business strategy and future growth.

To scale globally, HLL could advance into ***diagnostic and predictive analytics*** – answering the why of what is happening and what to do next. This transition will turn existing data into insight that informs decisions about hub locations, performance, learner engagement, and resource allocation etc.

## Current Metrics versus Data Analytics Tools

### HLL's LearnTools

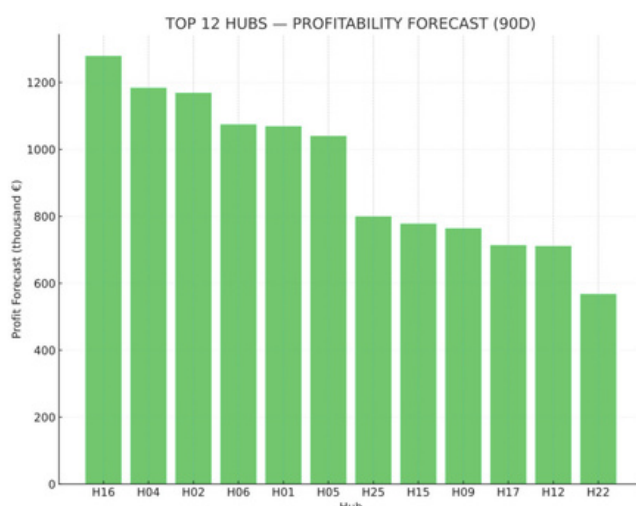
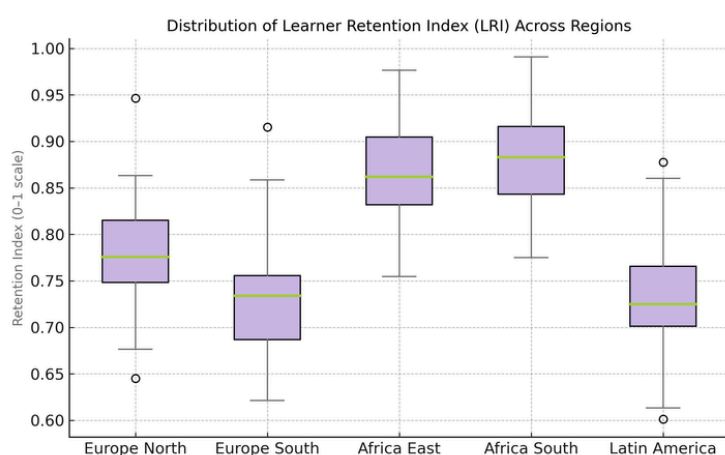
**LearnTools** is a digital platform designed to deliver course content, manage schedules, track goals, and support communication between learners and Learning Coaches. The focus is on improving student productivity and organization within HLL's hybrid learning model providing descriptive analysis rather than large-scale data analysis.

### What LearnTools Does

- **Schedule Management:** Enables learners to organize their own flexible timetables.
- **Goal Tracking:** Helps students and Coaches monitor student progress toward academic and personal goals.
- **Collaboration:** Facilitates communication between learners and Learning Coaches.
- **Productivity & Organization:** Provides a streamlined environment for learners to manage their workload and manages all aspects of the learning process from course design and delivery to tracking and reporting.

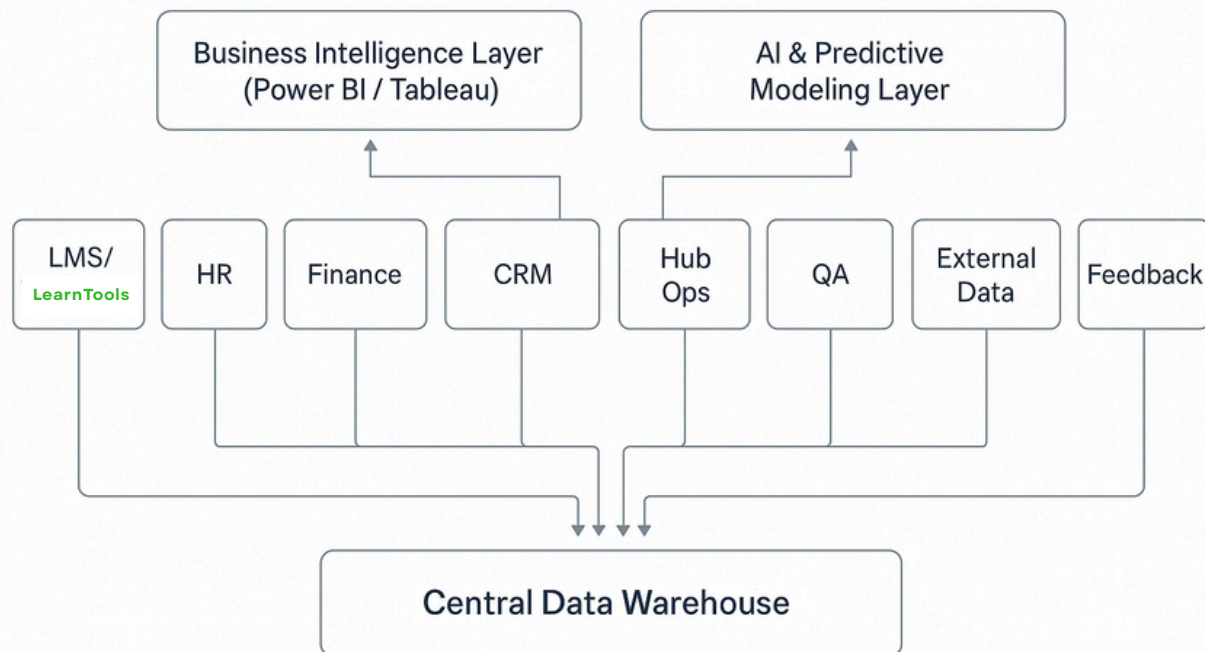
### Why It Is The First Layer of Data Analytics

- **Different Purpose:** LearnTools supports individual learning journeys. In-depth analytics tools are built for analyzing aggregated data to reveal trends and inform strategy.
- **Personal vs System-Wide:** LearnTools is learner-centered, whereas analytics systems integrate data across all areas: hubs, HR and finance, simultaneously to inform executive decisions.
- **Complementary Systems:** HLL's LMS dashboards offer progress tracking and excellent data sources, but comprehensive analytics would require connecting this data to further operational and strategic metrics.



## Integrated Data Architecture

Integrated data architecture would give Horizon Learning Labs (HLL) a unified, real-time view of its entire global network—connecting learning, operational, financial, and compliance data into one analytical framework. By breaking down silos between systems like **LearnTools**, HR, Finance, and Admissions, HLL could identify trends that are invisible in isolation—such as how coach stability affects learner retention or how engagement influences profitability. This architecture enables faster, evidence-based decisions, supports predictive modelling for growth and risk, and creates a foundation for eventual AI-driven insights that can guide expansion, resource allocation, and continuous improvement across all hubs.



### Quality Assurance as an example area of how analytics can merge compliance, HR and learner outcomes into a unified integrity metric:

- Curriculum compliance audits (whether hubs meet HLL’s academic standards)
- Accreditation reports from education authorities
- Internal quality checks (lesson reviews, documentation consistency)
- Learner and parent satisfaction surveys
- Incident reports or complaints

In analytics terms, QA data helps measure educational integrity, consistency, and risk. When merged with other data sources (like HR and Finance), it allows analysts to spot patterns – for example:

**“Hubs with lower QA scores also show higher coach turnover and lower student retention.”**

## Example Data Insight Dashboard Across Regions

Fictitious Data Comparison Across 25 Fictitious Hubs to find Regional Data Patterns and Trends

**Total Hubs: 25**

**Total Learners: 1125**

**Median WRI (Weighted Retention Index): 0.80**

**Median CSI (Coach Stability): 0.86**

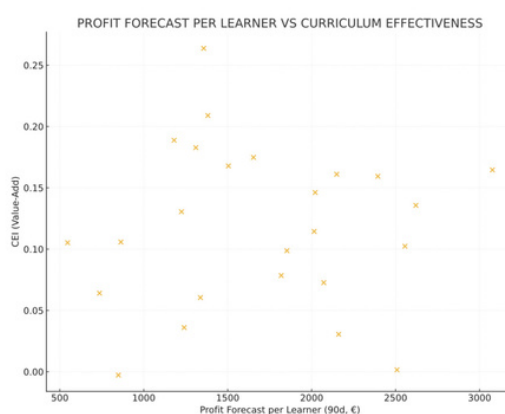
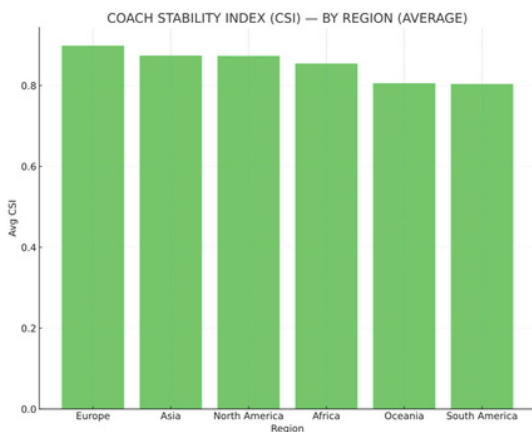
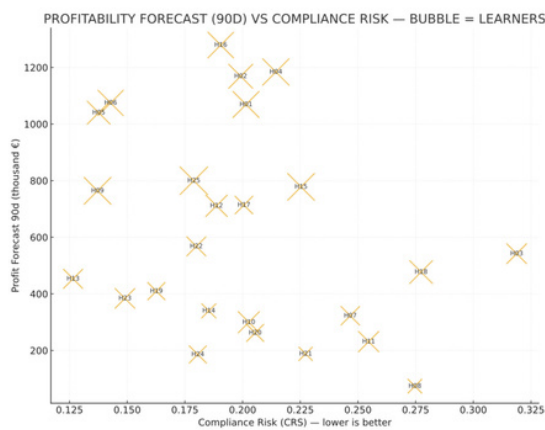
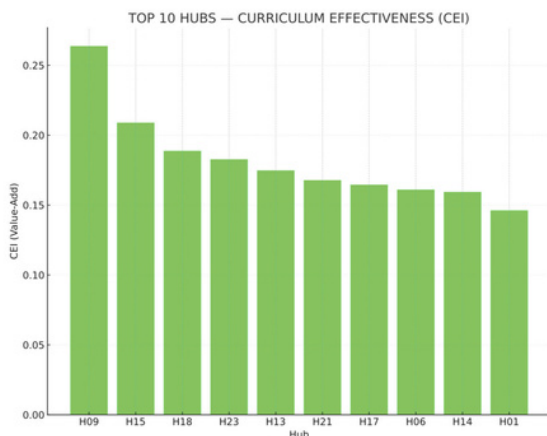
**Median CRS (Compliance Risk-lower is better) 0.20**

**90 day Profit Forecast \$15 318, 649**

**Median CEI (Curriculum Effectiveness) 0.11**

**Note: Metrics require data integration architecture to create a data warehouse that connects current data silos (LMS, Finance, HR and QA) beyond LearnTools and Excel etc. In a Power BI or Tableau driven data environment charts would:**

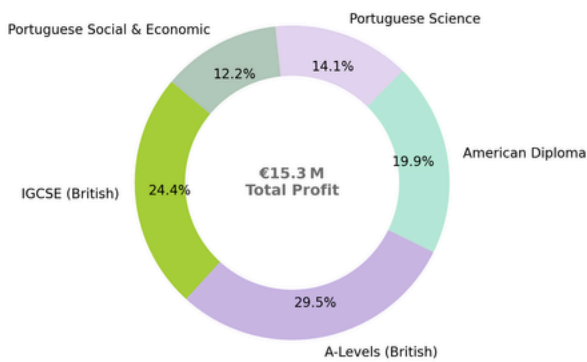
- update automatically,
- include DAX-driven KPIs (like weighted retention indices),
- feature drilldowns (hub → region → global).



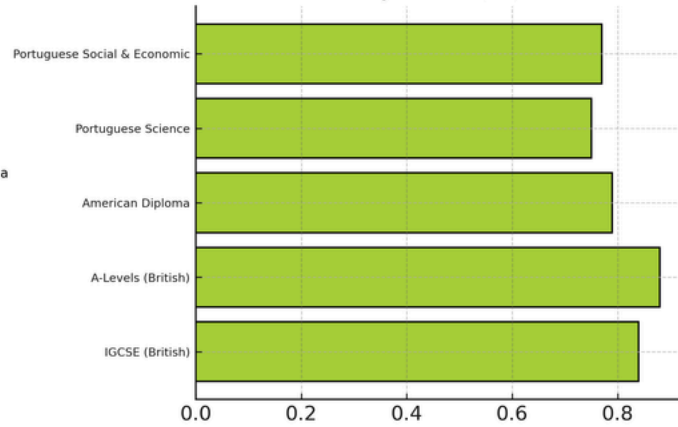
# Example Data Insight Dashboard Across Regions

## Data Comparison Across 25 Fictitious Hubs-Power BI Dashboard Mockup

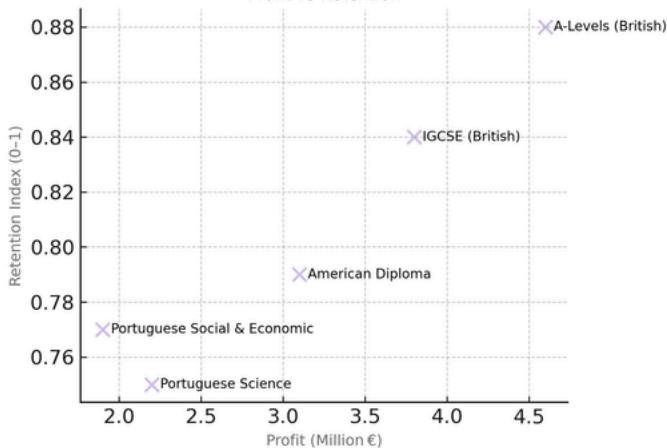
Profit by Curriculum



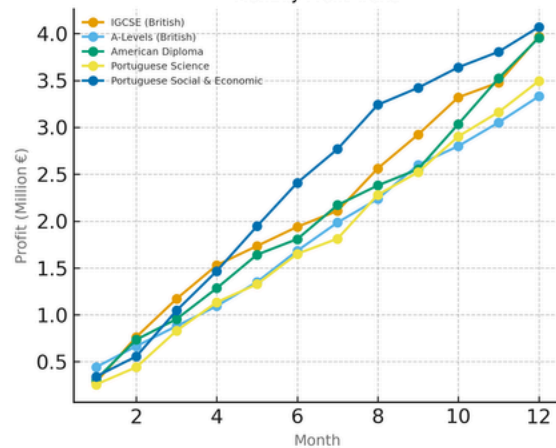
Avg Retention (LRI)



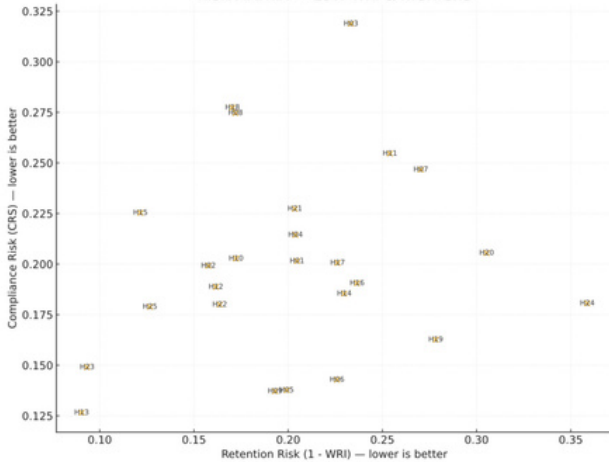
Profit vs Retention



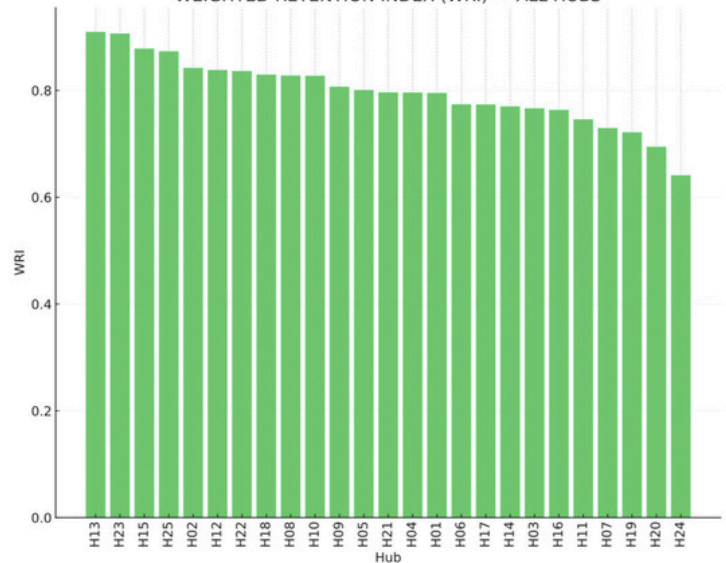
Monthly Profit Trend



RISK MATRIX — LOW WRI & HIGH CRS



WEIGHTED RETENTION INDEX (WRI) — ALL HUBS



# Top 4 (fictional) BI Questions for HLL in 2026

**Data Driven Forecasting Analytics Provide Key Insights:** These four questions reflect how cross-functional intelligence of a real data analytics backbone can build on what LearnTools or Excel dashboards can achieve.

## 1. Which hubs are driving long-term learner value versus short-term performance?

Why it matters:

**LearnTools** can show short-term grades or engagement, but it cannot identify where learners are retained, progressing, and converting into higher-level programmes. HLL needs to understand which hubs generate sustainable learner growth, not just weekly activity spikes.

### How data analytics helps:

- Integrates multi-year learner data with admissions and exit pathways.
- Builds a Learner Lifetime Value (LLV) metric that combines retention, referrals, and progression into new programmes.
- Uses time-series clustering to detect which hubs sustain academic and commercial value across multiple cohorts.
- Flags early-warning signals such as drop-off after two terms or low re-enrollment intent.

### Example insight:

*Nairobi hub generates the highest learner lifetime value (1.7× regional average) due to consistent coach continuity and 3-year retention of alumni."*

## 2. How does coach continuity and culture influence academic growth trajectories?

Why it matters:

**LearnTools** logs coach assignments but cannot reveal how coaching culture affects student development. HLL's model depends on relational continuity, which emerges only when HR, engagement, and assessment data are analysed together.

### How data analytics helps:

- Merges HR records (tenure, workload, satisfaction) with engagement and performance trends.
- Uses longitudinal correlation and causal modelling to map how coach stability influences learner growth, not just retention.
- Surfaces cultural signals such as peer feedback, coach sentiment, and workload balance.
- Identifies high-impact hubs where coaching practices drive exceptional progress.

### Example insight:

*"Hubs with more than 18-month average coach tenure show 0.4 standard deviations higher academic growth–stability outperforms coach-to-learner ratio as a success driver."*



## Top 4 (fictional) BI Questions for HLL in 2026 cont.

### 3. Which learning ecosystems (curriculum × hub × delivery × context) generate the best return on learning?

Why it matters:

**LearnTools** can track completion, but not the interaction between context and outcome. HLL operates across diverse economic, cultural, and pedagogical settings; to scale effectively, leadership needs evidence-based insight into where pedagogy meets profitability.

#### How data analytics helps:

- Combines curricular outcomes, local socioeconomic data, and cost models.
- Uses multivariate regression and factor analysis to isolate variables that drive high outcomes relative to cost.
- Calculates a Return on Learning Index (RoLI)—learning impact per euro spent.
- Recommends curriculum localisation strategies for each region.

Example insight:

*“A-level blended delivery in Namibia yields 22% higher Return on Learning than full-time attendance–equivalent outcomes at 18% lower cost per learner.”*

### 4. Which hubs and regions exhibit the strongest scalability resilience?

Why it matters:

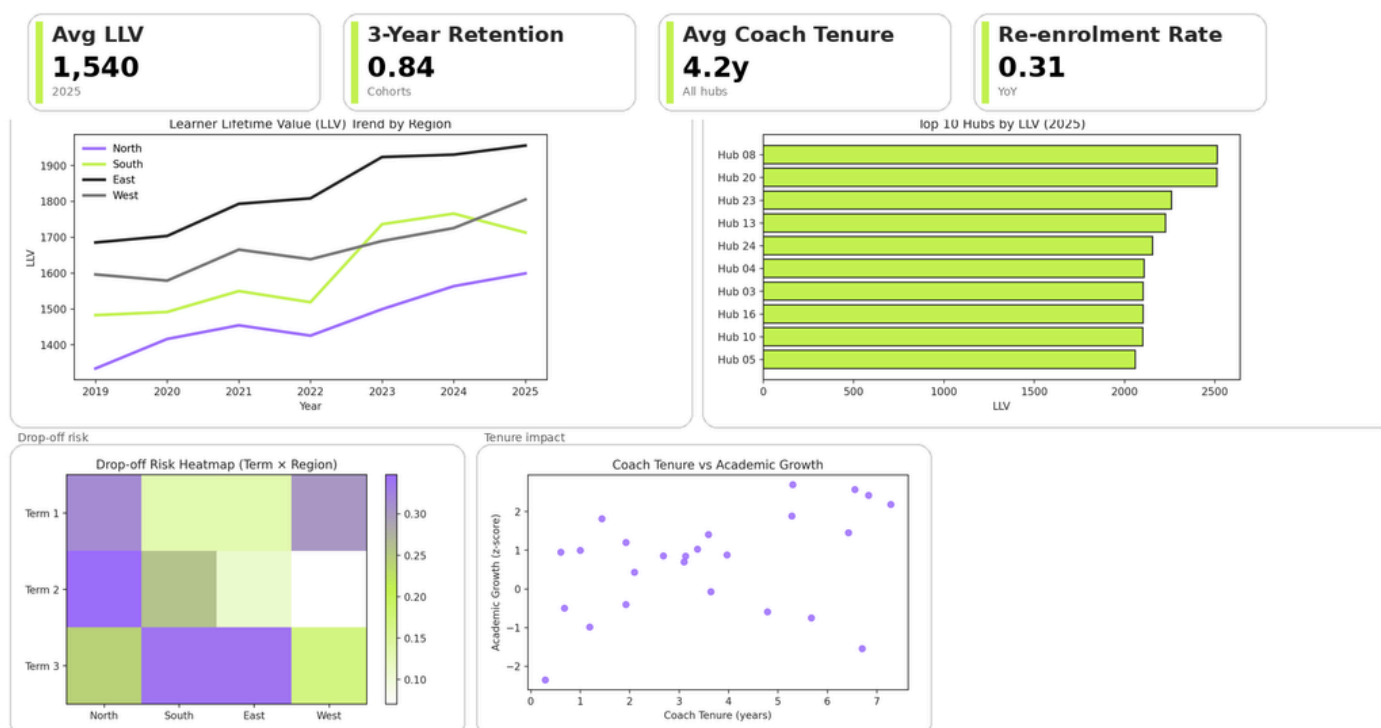
**LearnTools** reports enrollment and revenue but cannot forecast whether a hub model can scale without degrading quality or profitability. HLL’s franchise expansion requires predictive insight into operational elasticity—how fast a hub can grow before quality slips.

#### How data analytics helps:

- Integrates Finance, HR, and Learning data to compute scalability ratios: learner-per-coach load, rent per learner, QA variance, and satisfaction drift.
- Applies predictive modelling to simulate stress scenarios such as a 25% enrolment increase.
- Quantifies risk bands for each hub: resilient, fragile, or unsustainable.
- Supports decision trees for franchise replication.

Example insight:

*“Singapore South hub shows high profit per learner but low scalability resilience—QA variance exceeds 10% when enrolment grows beyond 150.”*



# Strategic Business Intelligence Domains for HLL

## Diagnostic and Predictive Data Analytics Provide Key Insights

### Human Capital and Efficiency

- Which coaches/hubs deliver the highest learner growth over time?
- Which staffing patterns (coach-to-learner ratios, tenure spread) optimize both profit and learning outcomes?
- Where is turnover likely in the next 90 days, and what early signals predict it?
- Are high-performing coaches clustered or evenly distributed across hubs?

### Operational Efficiency (Hubs & Processes)

- Which hubs are over/under capacity, and how does utilisation affect outcomes?
- What drives cost per learner variance by hub/region?
- Where do scheduling or staffing inefficiencies create bottlenecks?
- What's the average learner journey time (enrolment → certification) by hub?

### Curriculum & Learning Outcomes

- Which curricula/modalities (e.g., A-level, IGCSE, blended) yield best value-add by region?
- Which subjects show widest variance in outcomes, and why?
- How does digital engagement map to final assessment performance?
- Where are early indicators of learning decline or content misalignment?

### Financial Intelligence

- Which hubs have the strongest profit per learner after regional cost adjustments?
- How do pricing, rent, and staffing mix impact margin resilience?
- Which hubs are most likely to hit/exceed monthly/quarterly revenue targets?
- What is expected 12-month ROI for new hub openings under current assumptions?

### Quality & Compliance

- Which hubs carry the highest compliance risk and what are the root causes?
- How do QA audit scores correlate with coach turnover and learner outcomes?
- Which training interventions most effectively reduce repeat non-conformities?
- Where is accreditation readiness at risk in the next two terms



# Strategic Business Intelligence Domains for HLL

## Diagnostic and Predictive Data Analytics Provide Key Insights

### Learner Experience

- Which signals (logins, session time, sentiment) predict disengagement?
- Which hubs achieve the highest satisfaction and sense of belonging?
- Do specific coaching practices measurably improve persistence?
- Which journey touchpoints most influence re-enrolment and referrals?

### Predictive Modelling (Forecasts)

- Who is likely to churn next term (learner-level risk model)?
- Which hubs will miss retention, pass-rate, or margin targets without intervention?
- What enrolment/revenue can be expected by region under seasonality and pipeline?
- How will coach turnover shifts impact outcomes in the next 2-3 terms?

### Prescriptive Modelling (Actions)

- Which interventions (staffing, timetable, pricing, curriculum tweaks) maximise ROI per hub?
- What resource allocation yields the best improvement in retention next term?
- Which hubs should be replicated now vs. stabilised first, given risk/return?
- What is the optimal coach-to-learner ratio by context to balance cost and quality?

### Strategic Growth & Expansion

- Which cities/partners score highest on demand, unit economics, and delivery readiness?
- What hub maturity patterns predict scalable success vs. fragility?
- Where do demographic trends align with HLL's strongest programme fits?
- Which franchise candidates pass a data-driven readiness/quality screen?

